Natural frequencies and transient impact response of a 3D-printed carbon-fibre PETG fan housing, with an unmeasured elastic modulus

Clément Dumeril*

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Abstract

The housing of a desk fan printed in carbon-fibre PETG (PETG-CF) is analysed with code_aster for resonance with the rotor and for its impact response. Linear tetrahedra overestimate the first free-free modes by 12 to 20 %; quadratic tetrahedra on four meshes agree within 1 %, at 44.0, 64.3 and 109.9 Hz for $E = 6$ GPa. The modulus of the printed part is not measured. Between 3.8 GPa, the published value for printed PETG with 30 % fibre, and the 6.0 GPa of the original model, modes 3, 4 and 5 cross the third and fourth rotor harmonics and the blade-passing frequency at 2 000 rpm, so a resonance cannot be ruled out. Three impact cases (251 g, 2.5 ms pulse, contact patch held fixed) are run to 30 ms with linear transient dynamics and Rayleigh damping; verification found that the original oblique case fixed a slice through the housing, and it was replaced. With quadratic elements and a nominal 5 % damping, the peak von Mises stresses away from the contact, 114 to 174 MPa, sit at junctions of the arms and the stand and exceed the strength reported for printed PETG-CF. They are local (95th percentile below 23 MPa) and sensitive to the element order and to the simplified geometry of these junctions.

1 Introduction

Small rotating machines printed in fibre-reinforced polymers, such as desk fans, drone frames and robot covers, need two quick structural checks before a first print: whether a natural frequency of the printed part sits on an excitation line of the rotor, and where the part concentrates stress when it is knocked over. Both are routine with finite elements, but for a printed composite the inputs are uncertain. Fused-filament parts are anisotropic and weaker between layers [1], and short carbon fibres raise the stiffness by an amount that depends on fibre content and print orientation [2–4].

This study covers the housing of a desk fan designed at Polytechnia: a cylindrical shroud on a stand, printed in carbon-fibre PETG (PETG-CF), carrying a brushless motor in the shroud and the electronics in the base. The rotor has five blades and runs at a nominal speed of 2 000 rpm in this study. The contributions are:

- a free-free modal model of the housing with the motor and electronics as distributed masses, checked against the rotor harmonics, with the unmeasured Young's modulus carried as an uncertainty band rather than a single value;

*University of California, Berkeley (MEng). Work carried out in June 2026 at Polytechnia, an engineering consultancy co-founded by the author; convergence study and extended impact runs in October 2026. Repository: `https://github.com/clementdumeril/ventilator-modal-analysis-petgcf`.

- a mesh and element-order convergence study of the first modes, missing from the original analysis;
- a linear transient analysis of three impact cases, with damping, run past the end of the load pulse to the peak response, with stress metrics that separate the boundary-condition singularity at the contact from the stress in the housing, and a check of the contact patches that corrects one of the original cases.

2 Related work

The anisotropy of fused-deposition parts was characterised early by Ahn et al. [1]: strength and stiffness depend on raster orientation and drop across layers. Carbon-fibre reinforcement of printed polymers raises the in-plane stiffness when fibres align with the extrusion direction [2]. For PETG-CF printed by fused filament fabrication, Prabhakaran et al. [3] report a tensile modulus rising from 2.3 GPa for pure PETG to 3.8 GPa at 30 % fibre, with natural frequencies of test beams 18 % to 25 % higher; Batista et al. [4] and Kohn et al. [5] report that print parameters and build orientation change the mechanical response markedly, and that reinforcement does not always raise strength. The rotor-dynamic check used here is the Campbell diagram: frequencies of the structure plotted against rotor speed together with the lines $k\,\Omega$ of the rotor harmonics [6]; for a stationary housing the structural frequencies do not depend on speed. The mesh is generated with NETGEN [7] in SALOME [8]; modal and transient analyses use code_aster [9], with Newmark's average-acceleration scheme [10] for the transient runs.

3 Model

3.1 Geometry and mesh

The housing geometry comes from the Polytechnia FreeCAD model. The original CAD would not take a conforming tetrahedral mesh because of small faces and intersecting surfaces; it was defeatured (small faces and short edges removed, the detailed support arms replaced by simple ribs, one body removed from a fusion), which lowered the volume by 1.4 %, and oriented with the base on the xy plane and the height along $+z$. The reference mesh (Figure 1) has 95 358 nodes and 312 777 linear tetrahedra (TETRA4), from NETGEN with a minimum size of 1 mm, a maximum size of 10 mm and fineness 2.

3.2 Material and masses

PETG-CF is modelled as an isotropic linear-elastic solid (Table 1). The density is that of the filament. The Young's modulus of 6 GPa used in the original analysis has no documented source; the published value for FFF-printed PETG with 30 % carbon fibre is 3.8 GPa [3]. The results below are computed at 6 GPa and carried over the range 3.8 GPa to 6.0 GPa, whose upper bound is only the value of the original model. The motor (5010 brushless, estimated 0.090 kg) and the electronics (controller board, ESC and switch, estimated 0.075 kg) are not modelled as solids: their mass is added to two spherical groups of elements (radius 30 and 45 mm, elements with at least half of their nodes inside) by raising the density there, without adding stiffness. The total mass is 1.080 kg.

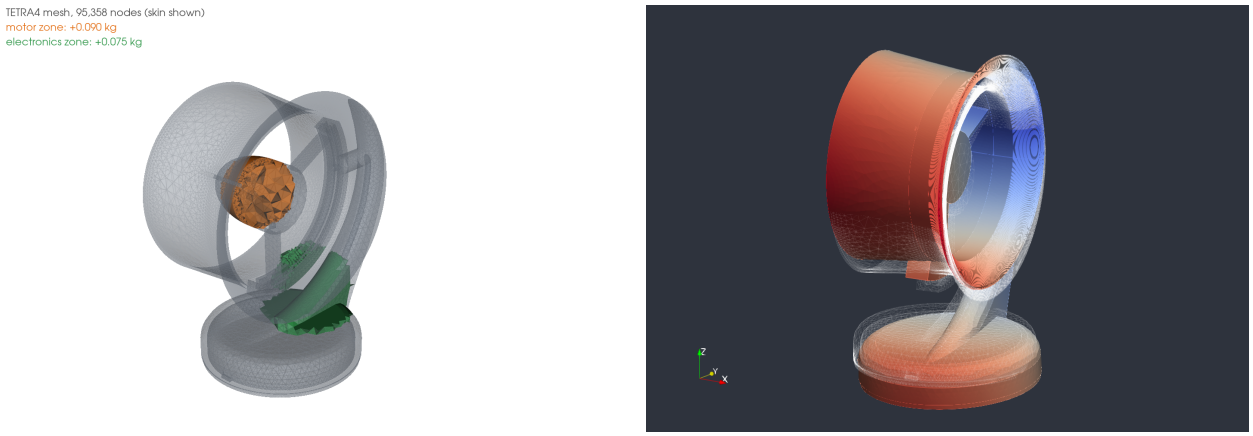


Figure 1: Left: reference mesh (skin shown) with the motor (orange) and electronics (green) mass zones. Right: first elastic mode at $E = 6\,\mathrm{GPa}$ (linear-element run of June 2026; deformation amplified).

Table 1: Model parameters.

Quantity	Value
PETG-CF	isotropic, $E = 6.0\,\mathrm{GPa}$ (range $3.8\,\mathrm{GPa}$ to $6.0\,\mathrm{GPa}$), $\nu = 0.35$, $\rho = 1\,300\,\mathrm{kg/m^3}$
Motor zone	sphere $R = 30\,\mathrm{mm}$, 23 127 elements, $+0.090\,\mathrm{kg}$
Electronics zone	sphere $R = 45\,\mathrm{mm}$, 11 616 elements, $+0.075\,\mathrm{kg}$
Total mass	$1.080\,446\,\mathrm{kg}$
Modal boundary conditions	free-free
Rotor	5 blades, nominal $2\,000\,\mathrm{rpm}$ ($n = 33.3\,\mathrm{Hz}$, BPF $= 166.7\,\mathrm{Hz}$)
Impact pulse	triangular, $2.524\,\mathrm{ms}$, peak $2.466 \times 10^4\,\mathrm{m/s^2}$ ($251\,\mathrm{g}$), $\Delta v = 3.11\,\mathrm{m/s}$

3.3 Modal problem

The free-free modes solve $\mathbf{K}\boldsymbol{\phi} = \omega^2\mathbf{M}\boldsymbol{\phi}$. The six rigid-body modes lie below 1 Hz; the search band starts at 1 Hz. For an isotropic material at fixed density and Poisson ratio, $\mathbf{K}$ is proportional to E and the mode shapes do not change, so $f(E) = f(E_0)\sqrt{E/E_0}$ exactly. Each mode is compared with the rotor lines $k\,n$ ($k = 1 \ldots 4$) and with the blade-passing frequency $5n$.

3.4 Impact model

The impact is idealised: the nodes of a small patch where the housing touches the floor are held fixed, and the rest of the housing receives a uniform acceleration pulse $a(t)$ (triangular, 2.524 ms, peak 251 g). The patch is the set of nodes within a tolerance of the floor plane, which is tangent to the housing at its extreme point in the impact direction. In the frame of the patch this is

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\,\mathbf{e}\,a(t), \tag{1}$$

with $\mathbf{e}$ the impact direction. The velocity change of the pulse is 3.11 m/s, the impact speed of a free fall of about 0.5 m with no rebound. There is no contact law, friction or rebound. Three directions are studied (Figure 5): lateral on the $-y$ side, where the shroud touches the floor (63 fixed nodes, of the linear mesh, tolerance 3 mm); lateral on the $+y$ side, where the stand touches first (63 nodes, 5 mm); and oblique, at 45° between $-y$ and $-z$, where the edge of the base touches first (107 nodes, 3 mm). The June study defined the oblique case differently; Section 5 explains why it was replaced.

4 Numerical method

Modes are computed with `CALC_MODES` in a frequency band (shift-invert Lanczos with a Sturm count check), code_aster 16.9 for the reference run and 17.4 for the convergence study. The impact runs of June 2026 used `DYNA_NON_LINE` with an elastic constitutive law up to 2.5 ms. Because the problem is linear, the October runs use `DYNA_VIBRA` in the physical basis with the same Newmark average-acceleration scheme ($\beta = 1/4$, $\gamma = 1/2$), $\Delta t = 0.1$ ms, up to 30 ms; the system is factorised once, which brings the cost from about 1 000 s to 66 s of CPU for the first 2.5 ms. Stresses are computed at the archived instants (every 0.2 ms). Because the fixed patch creates a stress singularity, the stress is reported as the maximum, 99th and 95th percentile of the element von Mises stress over the elements whose centroid lies more than 8 mm from the contact plane, next to the raw maximum.

5 Verification

Model checks. The chain was first run with steel properties: scaling the steel first mode (125.7 Hz) by $\sqrt{E/\rho}$ predicts 52 Hz for bare PETG-CF, against 49.2 Hz for the June linear-element model once the 0.165 kg of components is added. The mesh volume differs from the prepared CAD by 0.24 %. Recounting the mass-zone elements from the mesh file gives the same 23 127 and 11 616 elements as code_aster. The Sturm count finds 21 eigenvalues in 1 Hz to 1 500 Hz and 21 modes were computed, with residual norms between 2×10^{-11} and 4×10^{-8}. Rotating the model between orientations leaves mode 1 at 49.214 Hz.

Mesh and element order. The June analysis used a single mesh of linear tetrahedra. The convergence study (Table 2, Figure 2) remeshes the oriented geometry with NETGEN and changes the element order. Reducing the maximum element size from 10 to 5 mm barely changes the mesh (102 417 to 105 816 nodes), because the element size is set by the thin walls and the curvature, not by the maximum size; refinement was obtained through the NETGEN fineness instead (123 454 and 184 692 nodes). With linear elements the frequencies keep falling as the mesh is refined (mode 1 from 49.2 to 46.5 Hz, mode 3 from 131.9 to 122.3 Hz): linear tetrahedra are too stiff in the bending of thin walls. Quadratic elements (TETRA10) on four meshes from 92 381 to 123 454 nodes give mode 1 between 43.88 and 44.08 Hz, mode 2 between 63.95 and 64.52 Hz and mode 3 between 109.44 and 110.27 Hz, within 1 %. These meshes are close in size, so they show that the quadratic results are insensitive to the mesh rather than prove convergence; the spread is not monotonic with refinement and partly reflects the geometry each mesh captures (total mass 1.0794 to 1.081 4 kg). An independent estimate comes from the linear elements: a Richardson extrapolation of the two finest linear meshes, with order 2 in the element size $h \propto N^{-1/3}$, gives 43.6/64.2/111.9 Hz, within 1 to 2 % of the quadratic values. The first three frequencies are therefore taken as known to about 2 %, small next to the modulus band. The linear-element values of the June analysis are 12, 14 and 20 % too high for modes 1, 2 and 3, and the quadratic values on the reference mesh are used in Section 6. The extrapolation does not apply to modes 4 and 5, which are not in the asymptotic range with linear elements (Section 6).

Transient solver. The linear solver (`DYNA_VIBRA`) reproduces the June non-linear-solver results at every archived instant up to 2.5 ms (identical to four significant digits: 79.82 MPa, 95th percentile 7.070 MPa, displacement 4.238 mm at 2.5 ms for the $-y$ impact). Halving the time step to 0.05 ms on the same case over 15 ms changes the stress history by at most 0.4 % and leaves the peak unchanged (152.2 MPa at 10.8 ms; displacement 29.23 against 29.26 mm).

Table 2: Convergence of the first three elastic modes ($E = 6\,\mathrm{GPa}$). Nodes: corner nodes of the linear mesh. Last row: Richardson extrapolation of the two finest TETRA4 meshes.

Mesh, element	nodes (corner)	mass (kg)	f_1 (Hz)	f_2 (Hz)	f_3 (Hz)
published, TETRA4	95 358	1.0804	49.21	73.18	131.95
h10, TETRA4	102 417	1.0804	49.00	72.66	131.66
fineness 3, TETRA4	123 454	1.0814	47.37	69.80	125.49
fineness 4, TETRA4	184 692	1.0824	46.47	68.48	122.30
coarse (fineness 1), TETRA10	92 381	1.0794	43.88	63.95	109.44
published, TETRA10	95 358	1.0804	43.97	64.29	109.87
h10, TETRA10	102 417	1.0804	43.94	64.22	109.79
fineness 3, TETRA10	123 454	1.0814	44.08	64.52	110.27
TETRA4, Richardson ($p = 2$)	–	–	43.6	64.2	111.9

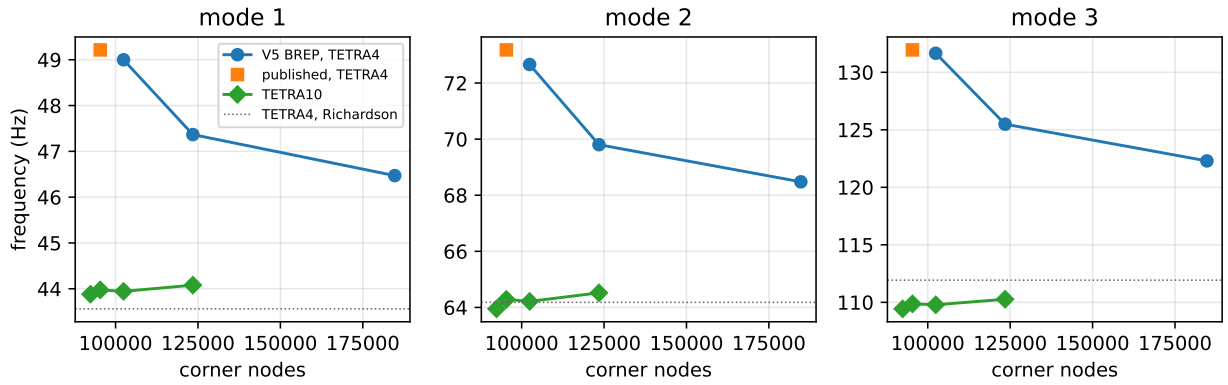


Figure 2: Frequencies of modes 1–3 against mesh size: linear tetrahedra (TETRA4) keep falling with refinement; quadratic tetrahedra (TETRA10) agree within 1 % on four meshes; dotted line: Richardson extrapolation of the two finest TETRA4 meshes.

Fixed patches. Plotting the fixed node groups on the mesh showed an error in the June oblique case. Its plane, with normal $(0, 1, 1)/\sqrt{2}$, was placed through the $-y$ side of the shroud and is not tangent to the housing: the 1 091 nodes within 1 mm of it form a slice through the shroud, the stand and the base, and 17 295 nodes lie beyond it. That case fixes a cut through the part rather than a contact patch, and its results (peak 91 MPa in June) are not used. A fall in that direction first touches the edge of the base, at the node that minimises $y + z$; the corrected oblique case puts the plane through that node. The planes of the two lateral cases are tangent, with no node beyond them (`shock/extended_2026-10/check_contact_planes.py` and its output).

Window length. Two cases with quadratic elements and $\zeta_n = 5\,\%$ had not settled at 30 ms and were continued to 60 ms: the $-y$ case, whose peak comes at 27.4 ms, and the oblique case, whose response beats. Between 26 and 30 ms both reproduce the 30 ms runs exactly. After 30 ms the $-y$ case reaches 163 MPa at 36.8 ms and at most 121 MPa from 40 to 60 ms, with the displacement decaying to 10.7 mm; the oblique case reaches at most 118 MPa, at 43.8 ms. The 30 ms window therefore contains the damped peaks.

6 Results

6.1 Modes against the rotor lines

With quadratic elements, which the convergence study supports (Section 5), the first five elastic modes at $E = 6\,\mathrm{GPa}$ are 44.0, 64.3, 109.9, 151.8 and 192.1 Hz (Figure 3); the published values, computed with linear elements, were 12 to 20 % higher for modes 1–3. In mode 1 the shroud moves on its stand while the base stays nearly still (Figure 1). At 6 GPa no mode sits on a rotor line at 2 000 rpm, but mode 2 is 3.6 % below the second harmonic.

Over 3.8 GPa to 6.0 GPa the modes scale by $\sqrt{E/E_0}$:

- mode 1 spans 35.0 Hz to 44.0 Hz: above the rotation frequency, but only by 5 % at 3.8 GPa; it would reach the first harmonic at $E = 3.45\,\mathrm{GPa}$, 9 % below the lower bound;
- mode 2 spans 51.2 Hz to 64.3 Hz, below the second harmonic over the whole range;
- mode 3 spans 87.4 Hz to 109.9 Hz and crosses the third harmonic (100 Hz) at $E = 4.97\,\mathrm{GPa}$;
- mode 4 spans 120.8 Hz to 151.8 Hz and crosses the fourth harmonic at $E = 4.63\,\mathrm{GPa}$;
- mode 5 spans 152.8 Hz to 192.1 Hz and crosses the blade-passing frequency at $E = 4.52\,\mathrm{GPa}$.

Between 4.5 and 5.0 GPa, within this range, three modes cross a rotor line. The crossings of modes 4 and 5 exist only with quadratic elements: with linear elements no mode lies between 132 and 228 Hz, even on the finest mesh. Their modes 4 and 5 are at 251 and 290 Hz on the reference mesh and still fall quickly with refinement (228 and 247 Hz on the finest mesh); they are probably the quadratic modes 4 and 5, overestimated by 50 to 65 %, but the mode shapes were not compared. Whether the printed housing has a resonance at 2 000 rpm therefore depends on a property that was not measured. Mounting adds a further shift: the free-free modes ignore the stiffness of whatever the base stands on. A mass-only estimate of the blades, $f\sqrt{M_0/(M_0 + \Delta M)}$ with $M_0 = 1.080\,\mathrm{kg}$, lowers every mode by 1 to 6 % for a total blade mass of 25 to 150 g; it ignores where that mass sits.

6.2 Impact

The June runs stopped at 2.5 ms, at the end of the pulse, while the stress was still rising in the two lateral cases. The October runs continue to 30 ms with linear elements at four damping levels ($\zeta_n = 0$, 2, 5 and 10 %) and with quadratic elements undamped and at the central level (Table 4; the corrected oblique case with quadratic elements at the central level only). The damping of the printed part is not known, and Rayleigh damping $\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K}$ is assumed, with α and β set to give a nominal ratio ζ_n of 2, 5 or 10 % at 15 and 300 Hz. Between these frequencies the ratio $\zeta(f) = \alpha/(4\pi f) + \pi\beta f$ is lower, down to $0.43\,\zeta_n$ at 67 Hz (Table 3). For $\zeta_n = 5\,\%$ it is 4.6 % at the first mode of the $-y$ case with the patch fixed (16.6 Hz), 2.4 % at that of the $+y$ case (42.6 Hz), and 2.1 to 2.9 % between 29 and 152 Hz. The labels 2 %, 5 % and 10 % below refer to ζ_n.

Undamped response. Without damping the housing keeps vibrating after the pulse, and the maximum over the window is set by when the modes happen to add up. With quadratic elements the $-y$ impact shows successive maxima of 171, 172, 179 and 220 MPa at 6.6, 13.6, 18.0 and 27.4 ms, and the displacement is still rising at 30 ms (39.8 mm): the undamped peak grows with the window and is not used as a result.

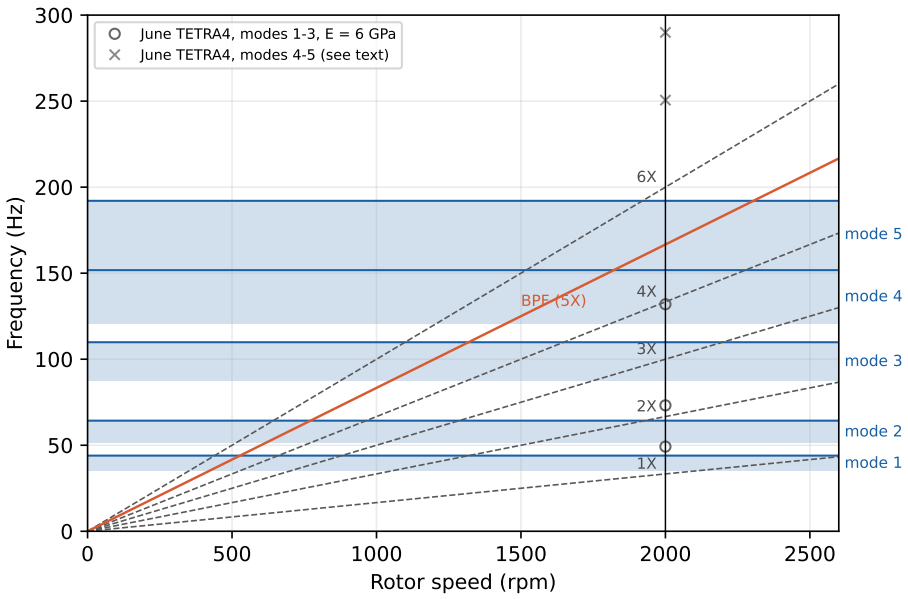


Figure 3: Campbell diagram with the TETRA10 modes of the reference mesh. Each band spans the frequency of a mode for $E = 3.8$ GPa (bottom) to 6.0 GPa (top, line). Open circles: June TETRA4 modes 1–3 at 6 GPa; crosses: June TETRA4 modes 4–5, probably the same modes as TETRA10 modes 4–5 (see text). Dashed lines: rotor harmonics; red line: blade-passing frequency.

Damped response. With damping the late maxima of the lateral cases disappear, and the peaks sit at junctions of the housing (Figure 5). For the $+y$ impact, where the stand touches the floor, the peak comes at the end of the pulse (2.6 to 3.0 ms) where the upper support arm meets the motor hub. For the $-y$ impact the housing rocks about the fixed patch (first frequency 16.6 Hz with the patch fixed, linear elements) and the peak comes after the pulse: at 10.8 ms with linear elements, where the stand joins the shroud on the side opposite the impact, and at 27.4 ms with quadratic elements, where the upper arm on the impact side meets the shroud. This late peak comes with the largest rocking displacement (34.7 mm at 29.8 ms): it is driven by the rotation of the whole housing about a patch that stays clamped, which a fan free to bounce or slide would not undergo in this form, and is probably an artefact of the boundary condition. For the oblique impact on the base edge the peak is again where the stand joins the shroud, at 6.2 ms with linear and 8.2 ms with quadratic elements. With quadratic elements this response beats, with a second maximum of 136 MPa at 25.6 ms; continued to 60 ms, its later maxima stay lower (at most 118 MPa, at 43.8 ms). Damping lowers the peaks slowly: from 152 to 128 MPa between $\zeta_n = 0$ and 10 % for the $-y$ impact with linear elements. Quadratic elements, which are less stiff in bending, give higher peaks at 5 %: 174 against 138 MPa for $-y$, 114 against 110 MPa for $+y$ and 139 against 112 MPa for the oblique impact. They also move the $-y$ peak to another junction, so the hot spots are best read as a set of regions (arm-to-hub, arm-to-shroud and stand-to-shroud junctions) rather than as single points. These peaks are mesh- and geometry-sensitive: the step from linear to quadratic elements raises them by 3 to 26 % (110 to 114, 112 to 139 and 138 to 174 MPa), and they sit at junctions where the detailed arms of the original CAD were replaced by plain ribs; where such a junction has a sharp re-entrant edge, the computed stress grows without bound as the mesh is refined. They are therefore not converged values. The 95th percentile of the stress outside the contact band stays below 26 MPa in all runs (23 MPa with damping), so the peaks are local. The raw maxima, which include the

Table 3: Damping ratio of the Rayleigh model, $\zeta(f)$, for the three nominal levels ζ_n (reached at 15 and 300 Hz). Patch-fixed modes: linear elements.

f (Hz)		ζ (%) for ζ_n = 2, 5, 10 %		
15.0	lower tuning frequency	2.0	5.0	10.0
16.6	mode 1 with the $-y$ patch fixed	1.8	4.6	9.1
42.6	mode 1 with the $+y$ patch fixed	0.9	2.4	4.7
44.0	free-free mode 1	0.9	2.3	4.6
67.1	minimum of the ratio	0.9	2.1	4.3
192.1	free-free mode 5	1.4	3.4	6.8
300.0	upper tuning frequency	2.0	5.0	10.0

Table 4: Impact peaks over 0 ms to 30 ms: von Mises stress σ (MPa) more than 8 mm from the contact plane, time of the peak t (ms) and height z (mm) of the peak element, for linear (TETRA4) and quadratic (TETRA10) elements and nominal Rayleigh ratio ζ_n (Table 3). Undamped peaks depend on the length of the window (see text).

		Lateral, Y−			Lateral, Y+			Oblique, base edge		
Element	ζ_n	σ	t	z	σ	t	z	σ	t	z
TETRA4	0 %	152	10.8	158	123	28.4	182	121	11.8	158
TETRA4	2 %	146	10.8	158	114	2.6	182	117	6.2	158
TETRA4	5 %	138	10.8	158	110	2.6	182	112	6.2	158
TETRA4	10 %	128	10.8	158	104	2.6	182	105	6.2	158
TETRA10	0 %	220	27.4	198	147	11.0	204	–	–	–
TETRA10	5 %	174	27.4	198	114	3.0	180	139	8.2	158

elements at the fixed nodes, reach 310 MPa; inside the contact band they are singularities of the boundary condition.

Comparison with the material strength. The tensile strength of FFF-printed PETG with 30 % carbon fibre is about 60 MPa in the plane of the layers [3]. For unfilled PETG, a manufacturer data sheet gives a tensile yield stress of 46 and 50 MPa for specimens printed flat and on edge, and a stress at break of 19 MPa for upright specimens, which measures the adhesion between layers [11]; Ahn et al. [1] and Kohn et al. [5] report the same weakness across layers. The damped peaks computed here, 114 to 174 MPa with quadratic elements at $\zeta_n = 5\,\%$, exceed the in-plane strength of PETG-CF by a factor of 2 to 3 and the interlayer strength by a factor of 6 to 9. Given how sensitive these peaks are, the comparison is qualitative. The computed peaks exceed the strength; whether the junctions would fail depends on what the idealisation leaves out: the patch stays clamped to the floor after the pulse instead of releasing, the pulse is a uniform 251 g on the whole housing, the junction geometry is simplified, and the material stays linear elastic. A linear model with local, mesh-sensitive peaks does not predict cracking. What it shows is that these junctions have no margin under a fall of this severity, which supports the direction of the June recommendation, extended to the hot spots found here: a soft bumper on the shroud and on the edge of the base, which lowers the peak deceleration by lengthening the stop, and larger fillets at the arm and stand junctions.

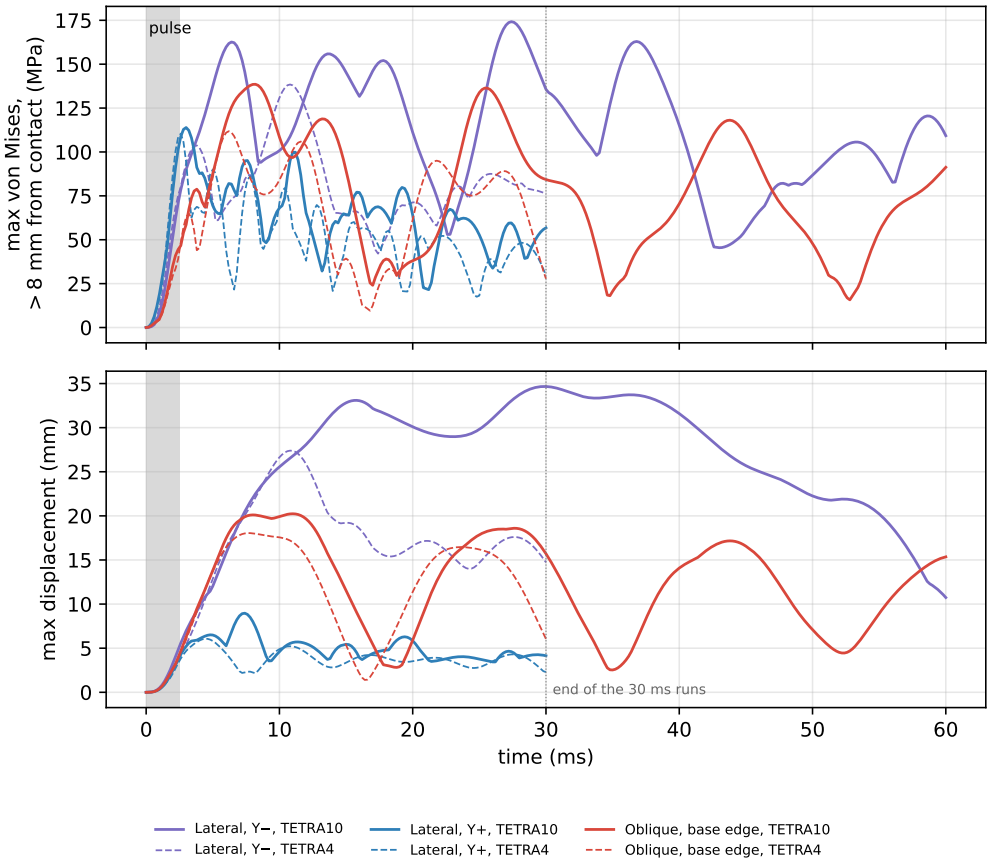


Figure 4: Impact histories at a nominal Rayleigh ratio $\zeta_n = 5\,\%$: maximum von Mises stress more than 8 mm from the contact plane (top) and maximum displacement (bottom), with quadratic (solid) and linear (dashed) elements. Grey band: the 2.5 ms pulse. Two cases with quadratic elements are continued to 60 ms.

7 Discussion and limitations

Two results of the June analysis do not hold. Its frequencies, from linear tetrahedra, were 12 to 20 % too high for the first three modes, so the closest call it reported (mode 3 near the fourth harmonic) is a different mode near the third harmonic once the element order is checked. Its impact stresses, read at 2.5 ms, stopped before the peak of the $-y$ case and used elements that are too stiff in bending. Its oblique case fixed a slice cut through the housing instead of a contact patch. What holds is that the stress concentrates locally, at a few junctions of the housing; the values of these local peaks are not converged.

The main uncertainty is the material. With E between 3.8 and 6.0 GPa, three modes cross a rotor line at 2 000 rpm; with an orthotropic printed material the mode shapes would also change, which the $\sqrt{E}$ scaling does not capture. The other limits are:

- *Material model.* Isotropic, linear elastic, no measured property; the printed part is anisotropic and weaker between layers [1, 5].
- *Geometry.* The detailed support arms were replaced by plain ribs to obtain a mesh; the stress hot spots are at these simplified junctions.
- *Boundary conditions.* Free-free for the modes; the stand, its support and any mounting are

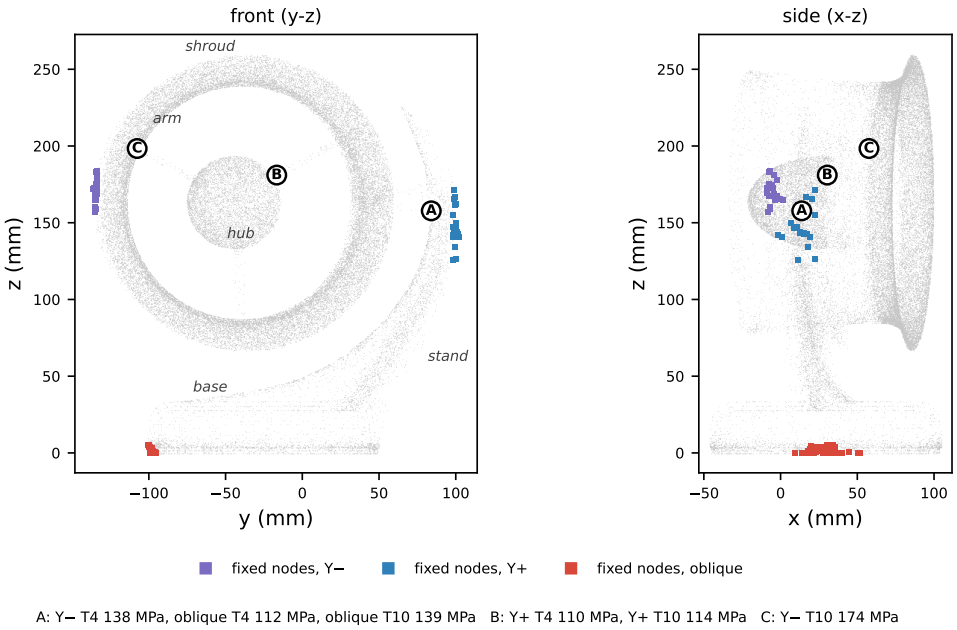


Figure 5: Fixed nodes of the three impact cases (coloured squares) and locations of the peak elements at $\zeta_n = 5\,\%$, labelled with the runs that peak there (T4: linear, T10: quadratic elements). A: junction of the stand with the shroud; B: junction of the upper arm with the hub; C: junction of the upper arm with the shroud, on the impact side of the $-y$ case. The outline is a random sample of the mesh nodes.

not modelled.

- *Masses.* Motor and electronics are estimated masses spread over spherical zones, with no added stiffness; the blades are not modelled.
- *Impact.* No contact law, friction or rebound: a patch of nodes stays fixed for the whole run. The pulse (251 g, 2.5 ms, $\Delta v = 3.11$ m/s) is uniform over the housing. Damping is Rayleigh with assumed ratios, lower between 15 and 300 Hz than the nominal value. The $-y$ case reaches displacements of 30 to 40 mm, at the edge of the small-displacement assumption.
- *Validation.* No tap test, accelerometer measurement or drop test.

8 Conclusion

The free-free modes of the PETG-CF housing, computed with quadratic elements and known to about 2 %, are 44.0, 64.3 and 109.9 Hz for the first three at $E = 6$ GPa, 12 to 20 % below the values of the original linear-element model. Carried over the modulus range 3.8 GPa to 6.0 GPa, modes 3, 4 and 5 each cross a rotor line at 2 000 rpm, so the absence of resonance cannot be claimed without measuring the material; the crossings of modes 4 and 5 appear only with quadratic elements. Under an idealised impact with the contact patch held fixed, the damped stress peaks at the arm and stand junctions (114 to 174 MPa with quadratic elements at a nominal 5 % damping) are local and sensitive to the mesh and the simplified geometry, and they exceed the strength reported for printed PETG-CF. In order of priority, the next steps are: a tap test of a printed housing to measure the first frequencies and infer the effective modulus; tensile tests of printed coupons in the layer

and across layers; a supported modal model with the stand on its real support; and a drop test of a printed housing with and without a bumper to calibrate the impact model.

Reproducibility statement

Inputs and scripts are in the repository: the oriented CAD (`cad/`), the meshing script (`src/mesh/mesh_carter_netgen.py`; SALOME 9.13, NETGEN, min 1 mm, max 10 mm, fineness 2), the reference modal input (`src/code_aster/`; code_aster 16.9), the convergence inputs and outputs (`data/convergence/`; code_aster 17.4), the June impact inputs (`shock/code_aster/`; code_aster 17.4) and the October impact generators, inputs, logs and time histories (`shock/extended_2026-10/`, `shock/results_extended/`; code_aster 17.4). All October runs used 2 threads on a 4-core Intel i7-6700HQ computer with 32 GB of memory: a TETRA4 modal run takes 3 to 5 min, a TETRA10 modal run 20 to 30 min and 11 to 18 GB, a 30 ms impact run 15 to 65 min (TETRA4) or 50 to 80 min (TETRA10), and a 60 ms TETRA10 run about 100 min with stresses from 26 ms on (with stresses at every archived instant, its scratch space exceeded the 118 GB free). Meshes and result files (4 GB to 6.5 GB per impact run) are not versioned. Tables and figures are regenerated by the scripts in `paper/scripts/`, and the paper by `paper/build_paper.sh`.

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